

Perturbation Semantics and Stability Separations for Local Learning Rules

Abstract

Algorithmic stability is not a single notion: it depends on which sample perturbation is applied and where the perturbed predictor is evaluated. We develop a systematic calculus of perturbation stability for deterministic k -nearest-neighbor classification. At the uniform level, we prove a decomposition theorem: replace-one stability is bounded by the sum of delete-one and insert-one stability, and the bound is algebraically sharp in the sense that neither operation alone suffices to control replacement perturbations. At the pointwise level, we construct, for every odd $k \geq 3$, an explicit finite metric sample for which all leave-one-out (LOO) stability indicators vanish while a single replace-one perturbation changes the prediction at a designated query. The same phenomenon occurs for even k under deterministic tie-breaking. These separations are not confined to pathological examples: they appear at measurable frequency in random graph benchmarks and UCI tabular datasets, and they are detectable by a margin-based diagnostic algorithm. An abstract local vote rule framework extends the margin-crossing criterion and the delete-plus-insert decomposition beyond k -NN. Together, the results establish that LOO stability and adversarial replace-one stability are structurally distinct even in the smallest finite local-rule setting, with implications for the interpretation of cross-validation diagnostics.

1 Introduction

Leave-one-out cross-validation is a standard tool for assessing classifier performance, and a low LOO error is often read as evidence that the classifier is not overly sensitive to any single training point. This inference conflates two distinct objects: the effect of deleting a point and then evaluating at that same point (LOO), versus the effect of replacing a point with an adversarially chosen labeled example (replace-one) and evaluating at an independent query. For local learning rules such as deterministic k -nearest neighbors, these two operations can give diametrically opposite answers about the stability of a prediction.

This paper is about making that distinction mathematically precise and showing that it matters for both theory and practice. Our setting is deterministic k -NN on finite metric spaces with ordered samples, duplicate occurrences, conflicting labels, and deterministic tie-breaking. Within this setting, we develop a perturbation calculus at two levels — a uniform (worst-case) level that relates perturbation operations algebraically, and a pointwise (fixed-sample) level that uncovers sharp separations between notions that are often conflated.

Perturbation operations. Let S be an ordered sample of labeled points. The three basic operations are: delete an occurrence (delete-one), insert a new labeled point (insert-one), and replace an occurrence by another labeled point (replace-one). Leave-one-out (LOO) is not a separate operation: it is delete-one evaluated at the deleted occurrence. This coupling between perturbation and evaluation point is the defining property that distinguishes LOO from replacement-based stability notions.

Uniform calculus. At the uniform (worst-case) level, the relationship between operations is clean. A replace-one perturbation factorizes into a delete-one followed by an insert-one at the same position. By the triangle inequality on losses, replace-one stability is bounded by the sum of delete-one and insert-one stability (Proposition 4.3). This bound is algebraically sharp: neither delete-one nor insert-one alone suffices to control replacement perturbations, as the separation theorems demonstrate.

Margin-crossing mechanism. For deterministic k -NN, the prediction changes if and only if the signed vote margin inside the ordered top- k neighborhood crosses zero (Proposition 5.1). This margin criterion is the unifying mechanism behind all perturbation analysis: every stability question reduces to tracking how the margin changes under an operation.

LOO/replace-one separations. The margin criterion reveals that LOO stability and replace-one stability can diverge sharply. We construct, for every odd $k \geq 3$ (and for even k under deterministic tie-breaking), an explicit finite metric sample in which all LOO indicators vanish while a single targeted replace-one perturbation flips the prediction at a designated query (Theorems 7.1, 8.1). The mechanism is the same in all cases: the margin at the LOO evaluation point is stabilized by duplicate occurrences, while a strategically placed label-flip at a different occurrence crosses the margin threshold.

Experimental evidence. These separations are not confined to hand-crafted witnesses. In random graph benchmarks and UCI tabular datasets, the LOO/replace-one gap appears at measurable frequency, especially for small k , high duplicate ratios, and label noise. We provide a margin-based diagnostic algorithm that detects vulnerable queries without enumerating all possible replacements.

Contributions

- (1) **Perturbation calculus.** We formalize delete-one, insert-one, replace-one, and LOO stability for deterministic k -NN with explicit conventions for ordered samples, duplicates, conflicting labels, and tie-breaking (Section 3). At the uniform level, replace-one stability is bounded by the sum of delete-one and insert-one stability (Section 4).
- (2) **Margin-crossing criterion.** We derive an exact signed-margin criterion for when a deterministic k -NN prediction changes after any perturbation (Section 5).
- (3) **LOO/replace-one separations.** We construct, for every odd $k \geq 3$ (and even k under deterministic tie-breaking), explicit finite metric samples in which all LOO indicators vanish while a targeted replace-one perturbation flips the prediction (Theorem 7.1, Theorem 8.1).
- (4) **Diagnostic and experiments.** We develop a margin-based diagnostic that detects vulnerable queries, and demonstrate that the LOO/replace-one gap occurs at measurable frequency in synthetic benchmarks and UCI datasets (Section 11).

Relation to Prior Work

Classical algorithmic stability [1, 13] studies how replacing one training example affects expected or fresh-test loss, primarily for generalization control. The present work is orthogonal: our goal is not a new generalization bound but a precise understanding of how perturbation operations differ at the level of individual finite samples. The deleted-estimate literature for nearest-neighbor rules [18, 7] and recent conformal prediction stability analyses [14, 16] touch on similar distinctions but do not isolate the LOO/replace-one separation we exhibit.

Organization

Section 3 fixes the notation and stability notions. Section 4 proves the uniform perturbation decomposition. Section 5 gives the signed-margin criterion. Section 6 states the 1-NN witness. Section 7 proves the general odd- k separation theorem. Section 8 treats the even- k case. Section 9 maps the hierarchy of stability relationships. Section 11 reports experimental results and the diagnostic algorithm. Section 12 discusses extensions to broader local vote rules. Section 13 discusses limitations and open problems.

2 Related Work

2.1 Algorithmic Stability and Generalization

The modern theory of algorithmic stability was formalized by Bousquet and Elisseeff [1], who defined several stability notions centered on replace-one perturbations and fresh-test evaluation, and used them to derive generalization bounds. Kutin and Niyogi [13] expanded the landscape with weaker and almost-everywhere variants, while Feldman and Vondrák [9] and later Bousquet, Klochkov, and Zhivotovskiy [2] obtained sharp high-probability bounds. Subsequent work connected stability to learnability and uniform convergence, clarifying when stability is not merely sufficient but also structurally related to learnability [19]. Mukherjee, Niyogi, Poggio, and Rifkin [15] further established that stability is sufficient for generalization and, for empirical risk minimization, necessary and sufficient for consistency. Hardt, Recht, and Singer [11] further showed that algorithmic stability explains generalization for optimization-based learning rules such as stochastic gradient descent. Our work is orthogonal to this line: we do not derive generalization bounds, but instead provide a microscopic perturbation calculus that clarifies which stability notion is being discussed in the first place. The classical literature mainly uses replace-one or delete-one with fresh-test evaluation; we show that LOO evaluation (common in practice) can give a qualitatively different picture.

2.2 Nearest-Neighbor Classification and Deleted Estimates

The nearest-neighbor line begins with Fix and Hodges [10] and the classical risk and consistency results of Cover and Hart [5, 4], together with the consistency framework formalized by Stone [20]. Those works study statistical performance under distributional sampling rather than worst-case single-perturbation effects. Closer to our setting are the deleted-estimate and local-rule analyses of Rogers and Wagner [18] and Devroye and Wagner [7, 8], which establish finite-sample bounds for local discrimination rules using leave-one-out style evaluation. These are important precedents, but their goal is to bound expected error, not to separate LOO from adversarial replace-one at the level of fixed finite metric witnesses.

2.3 Leave-One-Out and Conformal Stability

Kearns and Ron [12] connected algorithmic stability with leave-one-out cross-validation, already emphasizing that CV-style perturbation has its own logic. Recent conformal prediction work makes related distinctions operational, often in service of coverage certificates rather than 0–1 prediction loss. Examples include training-conditional guarantees [14, 16] and related stability analyses. Our results complement this line by showing that even the simpler 0–1 loss setting already exhibits a structural gap between LOO and replacement-based control, before any distributional or coverage considerations enter.

2.4 Finite Metric Witnesses and Local Learning Rules

Metric-space analyses of k -NN typically appear in consistency or geometric generality results [3] rather than perturbation-separation examples. To our knowledge, no prior work exhibits a finite graph-metric sample whose explicit purpose is to show that fixed-sample LOO stability need not control fixed-sample adversarial replace-one stability for deterministic k -NN under the conventions used here. The separation we prove is new in its specificity.

2.5 What the Paper Achieves

The paper’s distinctive contribution lies in combining (a) a systematic perturbation calculus that separates perturbation operations from evaluation points, (b) an exact uniform calculus relating those operations, (c) a general separation theorem valid for all odd $k \geq 3$ (and even k under tie-breaking), (d) an abstract framework extending beyond k -NN, and (e) reproducible experimental evidence that the separation occurs on real and synthetic data. Earlier computational search infrastructure [17] provided the 1-NN witness and candidate patterns for larger k , but left the general odd- k statement as a conjecture. The present paper upgrades that conjecture to a theorem.

3 Definitions and Preliminaries

3.1 Data Model and Sample

Definition 3.1 (Sample). A *sample* S of size n is an ordered tuple

$$S = ((x_1, y_1), \dots, (x_n, y_n))$$

where each $x_i \in X$ lies in a finite metric space (X, d) and each label $y_i \in \{0, 1\}$.

The order of the tuple is semantically relevant only for deterministic tie-breaking. Two identical labeled points appearing at different indices are treated as distinct sample occurrences.

Note 3.2 (Delete-One and LOO edge cases). The delete-one and LOO indicators are defined only for samples with at least two occurrences ($n \geq 2$). For $n = 1$, the $k' = \min(k, n - 1) = 0$, and the 0-NN predictor is undefined; we therefore exclude this case from stability analysis.

Note 3.3 (Duplicate points and conflicting labels). Duplicate points are allowed. Conflicting labels at the same point are allowed. Any theorem or experiment that excludes duplicates or conflicts must state this explicitly.

Careful separation of definitions. The conventions above are mathematically consequential. Ordered duplicates, conflicting labels, and deterministic tie-breaking are exactly the places where local instability can hide. Several later examples collapse or become ambiguous if these conventions are left implicit.

Definition 3.4 (Finite Metric Space). A finite metric space is a finite set X together with a distance function

$$d : X \times X \rightarrow \mathbb{R}_{\geq 0}$$

satisfying for all $x, x', z \in X$:

1. $d(x, x) = 0$;
2. $d(x, x') > 0$ for $x \neq x'$;
3. $d(x, x') = d(x', x)$;
4. $d(x, z) \leq d(x, x') + d(x', z)$.

Definition 3.5 (Graph Shortest-Path Metric). For a finite, simple, connected, unweighted, undirected graph $G = (V, E)$, the graph metric is

$$d_G(u, v) = \text{length of a shortest path from } u \text{ to } v.$$

Disconnected graphs are rejected in code paths that require a metric.

3.2 Prediction Rule

The query point x may be any element of X , including a point that appears in the training sample.

Note 3.6 (Deterministic Tie-Breaking). Neighbor ordering is determined lexicographically by:

1. smaller distance to the query point;
2. smaller sample index in the ordered tuple.

If class votes tie after the first k ordered neighbors are selected, the predicted label is resolved by fixed label order $0 \prec 1$. Ties are always broken in favor of label 0.

Definition 3.7 (k -NN Prediction). Given $k \in \mathbb{N}$ with $1 \leq k \leq n$, the deterministic k -NN predictor $h_S^{(k)}(x)$ is obtained by:

1. ordering the sample occurrences by the deterministic neighbor rule above;
2. selecting the first k occurrences;
3. taking the majority vote of their labels;
4. resolving label-vote ties by $0 \prec 1$.

3.3 Loss Function

Definition 3.8 (Binary 0-1 Loss). For a hypothesis h and a labeled point $(x, y) \in X \times \{0, 1\}$, the binary classification loss is

$$\ell(h, (x, y)) = \mathbf{1}\{h(x) \neq y\}.$$

3.4 Sample Perturbations

Let $S = ((x_1, y_1), \dots, (x_n, y_n))$.

Definition 3.9 (Delete-One). For index $i \in \{1, \dots, n\}$, define

$$S^{-i} = ((x_1, y_1), \dots, (x_{i-1}, y_{i-1}), (x_{i+1}, y_{i+1}), \dots, (x_n, y_n)).$$

Definition 3.10 (Insert-One). For any labeled point $z = (x, y) \in X \times \{0, 1\}$ and any insertion position $j \in \{1, \dots, n+1\}$, define

$$S \oplus_j z$$

as the ordered tuple obtained by inserting z at position j (shifting existing occurrences at indices $\geq j$ to the right by one).

Definition 3.11 (Replace-One). For index i and labeled point $z = (x', y') \in X \times \{0, 1\}$, define

$$S^{i \leftarrow z}$$

as the ordered tuple obtained by replacing the i -th occurrence of S with z .

In the worst-case stability definitions used here, the replacement point ranges over all labeled points in $X \times \{0, 1\}$, not over a data-generating distribution.

3.5 Stability Notions

Unless otherwise stated, all pointwise notions are evaluated with fixed k , fixed metric space, fixed tie-breaking rule, and fixed loss defined above.

Note 3.12 (Uniform stability sample-size convention). Pointwise stability statements concern a fixed training sample, a fixed perturbation, and a fixed evaluation pair (x, y) . Expected stability refers to the expectation under an explicitly specified data-generating distribution. Uniform stability is the supremum of the corresponding indicator over all valid ordered training tuples of a fixed size n , all allowed perturbation choices, and all evaluation pairs. We denote uniform delete-one and insert-one bounds by $\beta_{\text{del}}(n)$ and $\beta_{\text{ins}}(n)$ respectively.

Definition 3.13 (Pointwise Delete-One Stability Indicator). For index i and query-label pair (x, y) , define

$$\Delta_{\text{del}}(S, i, x, y) = \left| \ell(h_S^{(k)}, (x, y)) - \ell(h_{S^{-i}}^{(k')}, (x, y)) \right|,$$

where $k' = \min(k, n-1)$ when deletion reduces sample size below k .

Definition 3.14 (Pointwise Replace-One Stability Indicator). For index i , replacement z , and query-label pair (x, y) , define

$$\Delta_{\text{rep}}(S, i, z, x, y) = \left| \ell(h_S^{(k)}, (x, y)) - \ell(h_{S^{i \leftarrow z}}^{(k)}, (x, y)) \right|.$$

Definition 3.15 (Pointwise Insert-One Stability Indicator). For insertion position j and added labeled point z , define

$$\Delta_{\text{ins}}(S, j, z, x, y) = \left| \ell(h_S^{(k)}, (x, y)) - \ell(h_{S \oplus_j z}^{(k)}, (x, y)) \right|.$$

Definition 3.16 (LOO / CVloo Stability). For each index i , define

$$\Delta_{\text{loo}}(S, i) = \left| \ell(h_{S^{-i}}^{(k')}, (x_i, y_i)) - \ell(h_S^{(k)}, (x_i, y_i)) \right|.$$

LOO evaluates at the deleted sample occurrence itself. This definition is occurrence-based: if the same point appears twice, deleting one copy and evaluating at that occurrence is distinct from deleting the other.

For LOO, the uniform stability supremum is over ordered tuples S and indices i , with the evaluation pair fixed to the deleted occurrence.

Notion	Perturbation	Evaluated at	Supremum over	Distribution?	Used for
delete-one	remove one occurrence	arbitrary query-label pair	indices and queries	no	uniform and pointwise perturbation analysis
replace-one	swap one occurrence for adversarial labeled point	arbitrary query-label pair	indices, replacements, queries	no	strong local perturbation control
insert-one	insert one occurrence at position j	arbitrary query-label pair	insertion positions, added points, queries	no	decomposition of replace-one
LOO	delete one occurrence	that deleted occurrence	delete index only	no	cross-validation style local stability
uniform stability	usually replace-one in classical literature	fresh test point or worst-case query	all samples	often yes	generalization bounds

Table 1: Comparison of perturbation notions used in this paper and in the surrounding stability literature. Computational witness sections use fixed-sample worst-case indicators, not distributional expected stability.

3.6 Consistency Note

The consistency notion used in paper-level discussion is classical distributional consistency for k -NN classification. It is not a finite-sequence surrogate notion. Any finite-metric experiment provides evidence about worst-case stability behavior; it does not by itself constitute a consistency statement.

4 Uniform Stability Calculus

At the level of perturbation operations, replace-one is the composite move obtained by deleting an occurrence and then inserting a new labeled point. That observation is almost tautological at the sample level, but it is worth writing down explicitly because it yields the correct direction of implication for worst-case stability control.

Note 4.1 (Insert-One Operation). For an ordered tuple T of size m and any labeled point $z \in X \times \{0, 1\}$, the result of inserting z at position $j \in \{1, \dots, m+1\}$ is denoted $T \oplus_j z$. The notation $S^{i \leftarrow z} = (S^{-i}) \oplus_i z$ means: first delete occurrence i from S , obtaining S^{-i} , then insert z at the original position i of the deleted occurrence. This preserves the ordered structure and aligns with the in-place replacement semantics.

Note 4.2 (Uniform insert-one stability). When we speak of bounded insert-one stability $\beta_{\text{ins}}(n)$ uniformly over all insertion positions, we mean: for every ordered sample T of size m , every insertion position $j \in \{1, \dots, m+1\}$, every labeled point z , and every query-label pair (x, y) ,

$$|\ell(A(T), (x, y)) - \ell(A(T \oplus_j z), (x, y))| \leq \beta_{\text{ins}}(m).$$

The bound may depend on the sample size m before insertion.

Proposition 4.3 (Uniform replace-one via delete-and-insert). *Fix a deterministic learning rule A that maps ordered samples to predictors and a bounded loss $\ell \in [0, 1]$. Suppose that:*

- (1) *Delete-one stability is bounded by $\beta_{\text{del}}(n)$ uniformly over all ordered samples of size n , all indices, and all query-label pairs.*
- (2) *Insert-one stability is bounded by $\beta_{\text{ins}}(n)$ uniformly over all ordered samples of size n , all insertion positions j , all admissible labeled points z , and all query-label pairs.*

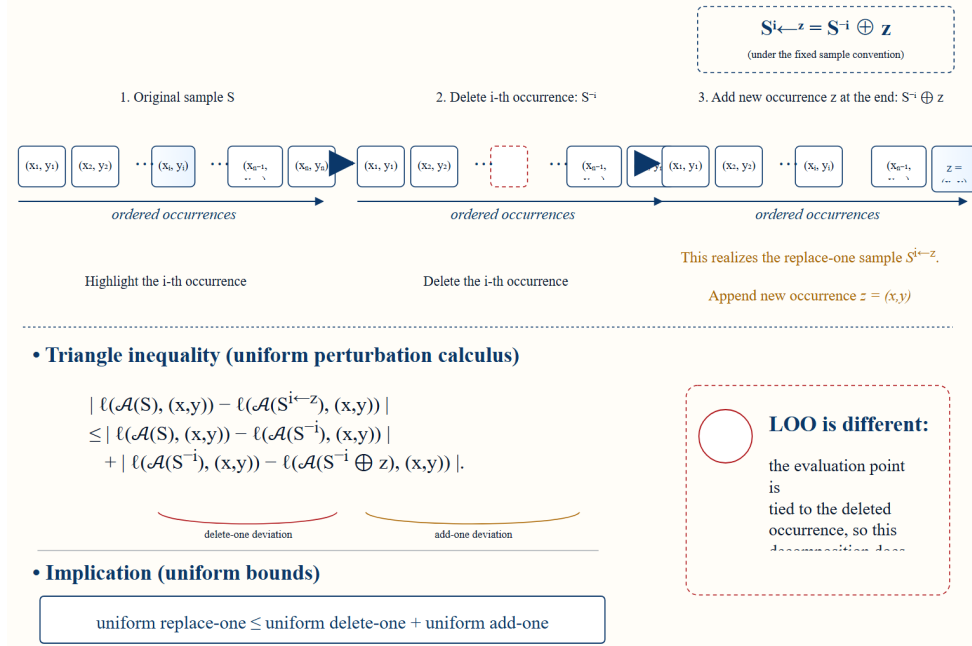


Figure 1: Operation-level factorization of a replace-one perturbation into a delete-one step followed by an insert-one step. The proof of [Theorem 4.3](#) is just the triangle inequality on the corresponding binary losses.

Then every replace-one perturbation satisfies

$$| \ell(A(S), (x, y)) - \ell(A(S^{i \leftarrow z}), (x, y)) | \leq \beta_{\text{del}}(n) + \beta_{\text{ins}}(n-1).$$

Note 4.4 (Proof sketch for [Theorem 4.3](#)). For fixed $S, i, z, (x, y)$, let $T = S^{-i}$. By the insertion convention, $S^{i \leftarrow z} = (S^{-i}) \oplus_i z$. Therefore

$$\begin{aligned}
| \ell(A(S), (x, y)) - \ell(A(S^{i \leftarrow z}), (x, y)) | & \leq | \ell(A(S), (x, y)) - \ell(A(T), (x, y)) | \\
& \quad + | \ell(A(T), (x, y)) - \ell(A(T \oplus_i z), (x, y)) | \\
& \leq \beta_{\text{del}}(n) + \beta_{\text{ins}}(n-1)
\end{aligned}$$

by the triangle inequality and the uniform assumptions. The last line uses that S^{-i} has size $n-1$, so the insert-one bound applies with that sample size.

[Theorem 4.3](#) is deliberately abstract. It is not specific to k -NN, and it does not yet tell us whether the bound is informative for a particular local learner. Its role is to freeze the algebra of perturbation types before we zoom in on the exact combinatorics of deterministic nearest neighbors.

The proposition also clarifies what it does *not* say. A pointwise LOO indicator cannot simply be dropped into the role of delete-one control above, because LOO changes the evaluation point along with the sample. Similarly, a fresh-test or expected-loss stability statement from the classical literature does not automatically transfer to the worst-case fixed-query indicators used in the present paper. The whole point of the calculus is to prevent these invalid transfers between notions.

5 Exact Deterministic k -NN Perturbation Criterion

Once the perturbation type is fixed, a deterministic k -NN prediction can only change for one structural reason: the signed class vote inside the ordered top- k neighborhood crosses zero. This section makes that statement exact.

For an ordered sample S and query x , let $\pi_S^x(1), \dots, \pi_S^x(n)$ be the sample indices sorted by increasing distance to x and then by sample index. Write

$$N_k(S, x) = (\pi_S^x(1), \dots, \pi_S^x(k))$$

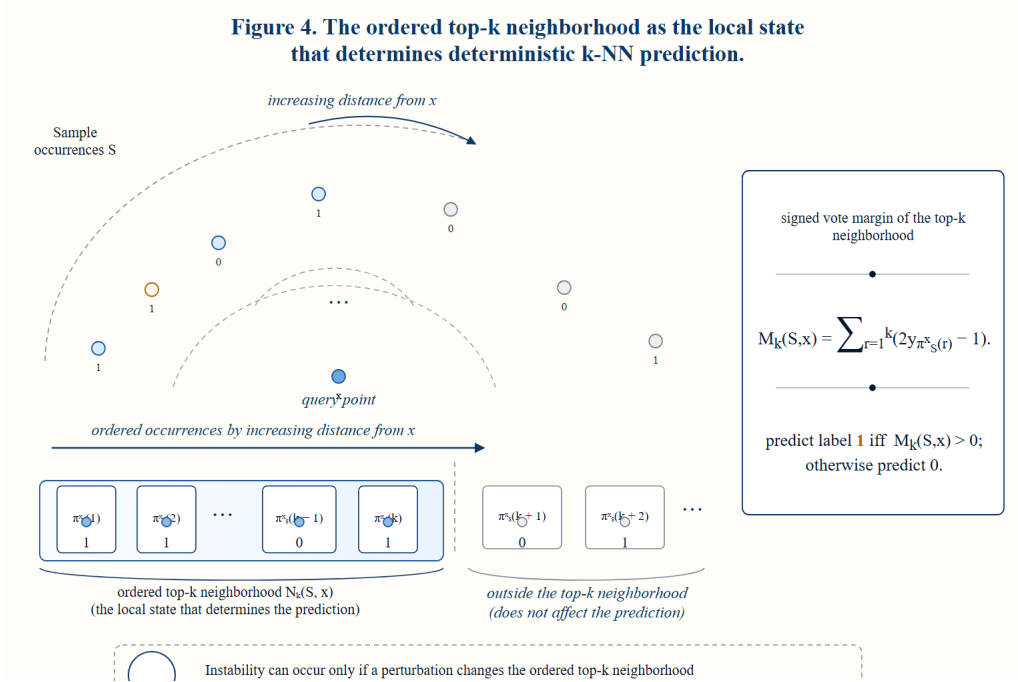


Figure 2: Ordered top- k membership determines the relevant local state for deterministic k -NN. Instability is possible only when a perturbation changes this neighborhood or flips the labels of occurrences already inside it.

for the ordered top- k neighbor list and define the signed vote margin

$$M_k(S, x) = \sum_{r=1}^k (2y_{\pi_S^x(r)} - 1).$$

Under the label-order tie-break $0 \prec 1$, the deterministic classifier predicts label 1 exactly when $M_k(S, x) > 0$.

Proposition 5.1 (Margin crossing criterion). *Fix S , k , and a query x . Let S' be any perturbed sample of the same ambient metric space, interpreted with the same deterministic tie-breaking rule. Then*

$$h_S^{(k)}(x) \neq h_{S'}^{(k)}(x)$$

if and only if the signed vote margins satisfy

$$(M_k(S, x) \leq 0 < M_k(S', x)) \quad \text{or} \quad (M_k(S', x) \leq 0 < M_k(S, x)).$$

Proof. By definition of the deterministic vote rule, the prediction is 1 exactly for strictly positive signed margin and 0 otherwise. Therefore the prediction changes precisely when one margin is positive and the other is non-positive. $\square$

Theorem 5.1 is elementary but useful: every perturbation argument can be reduced to tracking how the ordered top- k membership changes and how the corresponding signed vote changes.

Two immediate consequences are worth isolating. First, deleting an occurrence outside $N_k(S, x)$ cannot change the prediction at x unless the deletion also shifts the boundary and pulls a new occurrence into the top- k set. Second, a replace-one perturbation can act in two different ways: it can change the geometry of who enters the neighborhood, or it can leave the geometry fixed and still flip the vote by changing the label attached to a strategically placed occurrence.

These mechanisms become especially sensitive in the presence of duplicate occurrences and equal-distance contenders, because the top- k neighborhood is then controlled by the deterministic tie-breaking convention rather than by distance alone. That is not a nuisance artifact. It is part of the object being studied.

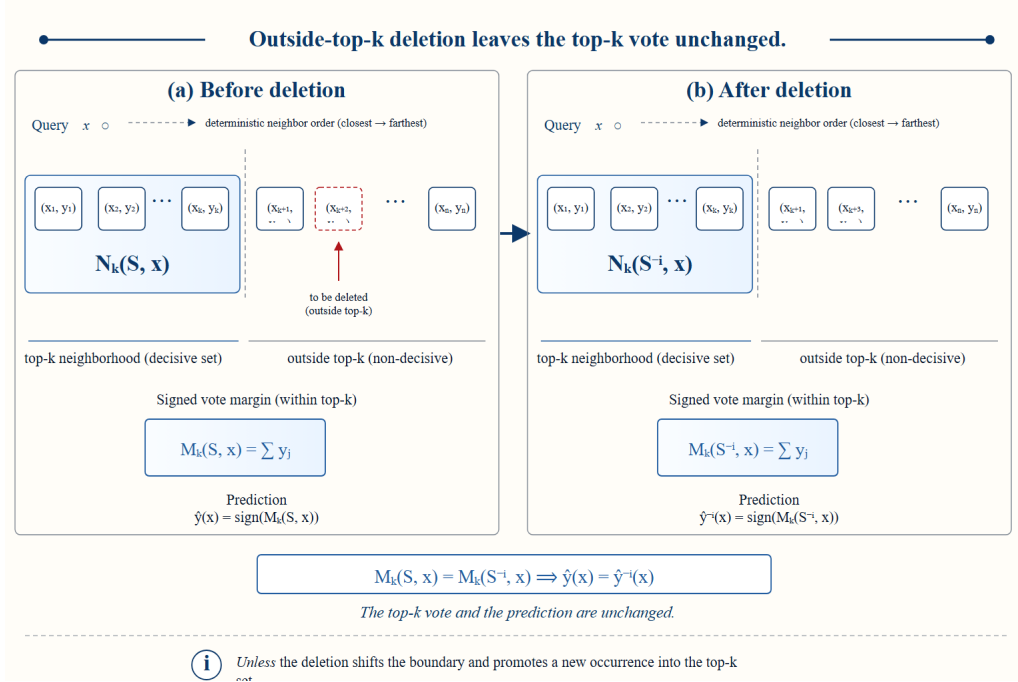


Figure 3: Two cases for deletion. Case A (left): deleting an occurrence outside the ordered top- k neighborhood is inert; the top- k set and margin are unchanged. Case B (right): deleting an occurrence inside the top- k set may promote the rank- $(k + 1)$ occurrence into the vote, changing the margin.

The rest of the paper repeatedly uses this margin-based analysis. In the 1-NN witness of [Section 6](#), the decisive effect is a single nearest-occurrence replacement. In the odd- k separation theorem of [Section 7](#), the construction engineers repeated occurrence patterns whose signed margin is stable under LOO but fragile under a targeted replace-one move.

6 A Two-Point Separation Between LOO and Replace-One Stability

We now come to the main explicit witness. The statement is small, explicit, and completely deterministic.

Note 6.1 (Remark on the example). The example uses no distance tie at the decisive replace-one query; the separation is therefore not merely an artifact of nearest-neighbor tie-breaking.

Proposition 6.2 (Two-point 1-NN separation). *There exists a finite graph metric space, an ordered labeled sample S , and a query-label pair (x, y) such that*

$$\Delta_{\text{loo}}(S, i) = 0 \quad \text{for every sample index } i,$$

while for some index j and some admissible replacement point z ,

$$\Delta_{\text{rep}}(S, j, z, x, y) = 1.$$

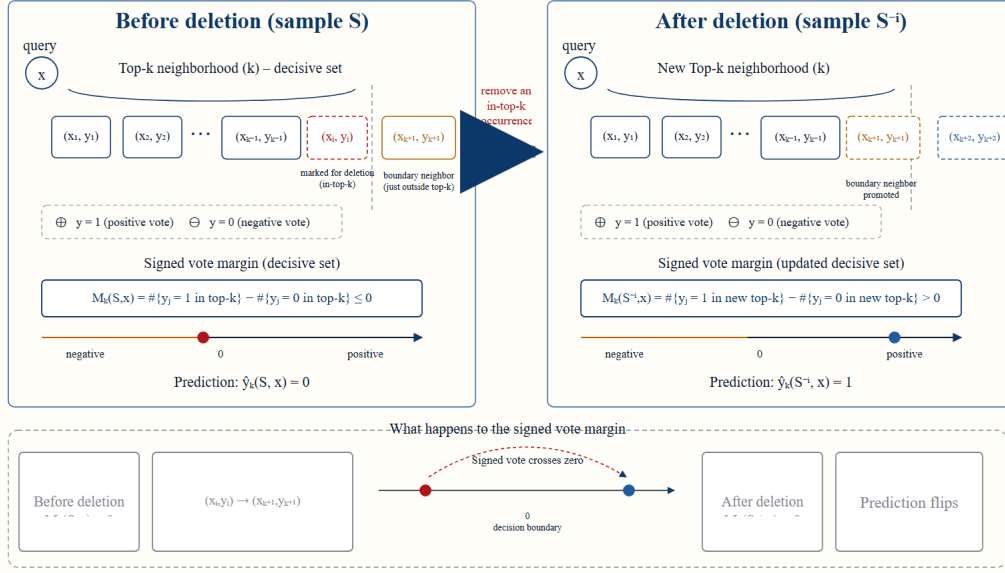
Note 6.3 (Proof sketch). Let the metric space contain two vertices a, b joined by one edge, so $d(a, b) = 1$. Consider the ordered sample

$$S = ((a, 0), (b, 0)).$$

Use deterministic 1-NN with the fixed tie-breaking conventions from [Section 3](#).

For LOO, delete the first occurrence. The remaining sample is $((b, 0))$, whose 1-NN prediction at the deleted occurrence's location a is still 0 because the sole remaining neighbor has label 0. Hence the loss

Figure 6. Instability Mechanism: Deleting an In-Top-k Occurrence Exposes a New Boundary Neighbor and Causes the Signed Vote Margin to Cross Zero



Deleting an in-top-k occurrence can expose a new boundary neighbor, change the signed vote margin's sign

Figure 4: A local perturbation changes the prediction only when the top- k signed vote crosses zero. This is the exact mechanism behind all explicit witnesses in this paper.

at $(a, 0)$ is unchanged. The same argument applies when deleting the second occurrence and evaluating at $(b, 0)$. Therefore $\Delta_{\text{loo}}(S, 1) = \Delta_{\text{loo}}(S, 2) = 0$.

Now evaluate at the fixed query-label pair $(x, y) = (a, 0)$. Replace the first occurrence $(a, 0)$ by $(a, 1)$, obtaining

$$S^{1 \leftarrow (a, 1)} = ((a, 1), (b, 0)).$$

At query a , the original sample predicts 0 because $(a, 0)$ is the unique nearest occurrence. In the perturbed sample, the unique nearest occurrence is now $(a, 1)$, so the prediction becomes 1. Hence $\Delta_{\text{rep}}(S, 1, (a, 1), a, 0) = 1$.

Theorem 6.2 gives the same witness in explicit form. What matters is not that the example is exotic, but that it is simple enough to inspect line-by-line. The phenomenon already appears before any asymptotics, averages, or large combinatorial constructions enter the story.

The supplementary material contains a finite-search pipeline that looks for smaller or structurally simpler witnesses under various restrictions. The witness in **Theorem 6.2** is theorem-level because it is explicit and hand-checkable. Claims that no witness exists below a certain search frontier are *not* theorem-level; they are computational certificates tied to the enumerated search space.

This distinction is worth making explicit. A paper can responsibly say “here is an explicit witness we know” and separately say “here is a reproducible finite certificate supporting minimality within the searched regime.” It should not blur those two statements together.

7 Odd- k Separation: A General Construction

The two-point 1-NN witness of [Section 6](#) shows that LOO stability and replace-one stability are not interchangeable for $k = 1$. A natural question is whether the same phenomenon persists for larger odd k . We answer this question in the affirmative by constructing, for every odd $k \geq 3$, a deterministic finite metric sample in which all LOO indicators vanish while a targeted replace-one perturbation changes the loss at a designated query-label pair. (The $k = 1$ case is already covered by the two-point witness of [Section 6](#).)

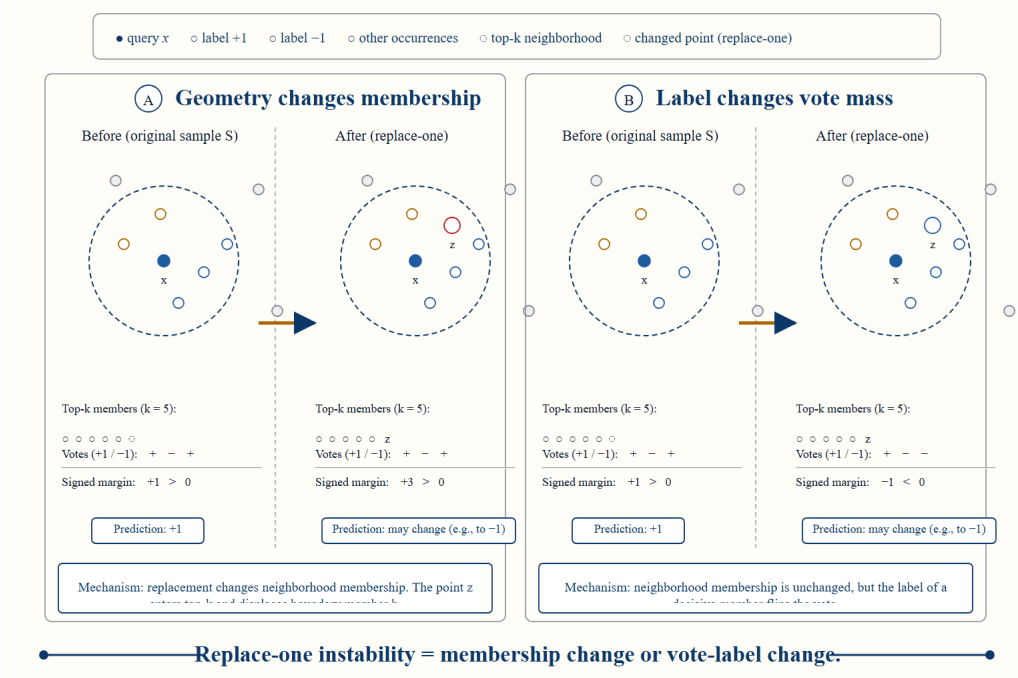


Figure 5: Replace-one perturbations can flip the decision either by moving a new occurrence into the decisive neighborhood or by changing the label composition of a neighborhood that was already geometrically close to the boundary.

7.1 Construction

Let $k = 2m + 1$ for an integer $m \geq 1$. Let the metric space consist of two distinct points a and b with $d(a, b) > 0$; any metric space with at least two points suffices. (For concreteness, one may take the two-vertex graph metric with $d(a, b) = 1$.)

Define the ordered sample

$$S_m = ((a, 0)_1, \dots, (a, 0)_{m+1}, (a, 1)_1, \dots, (a, 1)_m, (b, 0)_1, \dots, (b, 0)_m),$$

where subscripts track the occurrence index within the ordered tuple. The sample has size $n = (m + 1) + m + m = 3m + 1$.

Fix the query-label pair $(x, y) = (a, 0)$ and the neighborhood size $k = 2m + 1$.

Theorem 7.1 (Odd- k LOO/replace-one separation). *For every odd $k = 2m + 1$ with $m \geq 1$, the sample S_m satisfies*

$$\Delta_{\text{loo}}(S_m, i) = 0 \quad \text{for every index } i,$$

while there exists an index j and a replacement point z such that

$$\Delta_{\text{rep}}(S_m, j, z, a, 0) = 1.$$

Thus, pointwise LOO stability does not imply pointwise replace-one stability for deterministic k -NN with odd $k \geq 3$.

Proof. We analyze the margin behavior at the query a and at each deleted occurrence. Recall from [Theorem 5.1](#) that a deterministic k -NN prediction changes if and only if the signed vote margin $M_k(S, x)$ crosses zero.

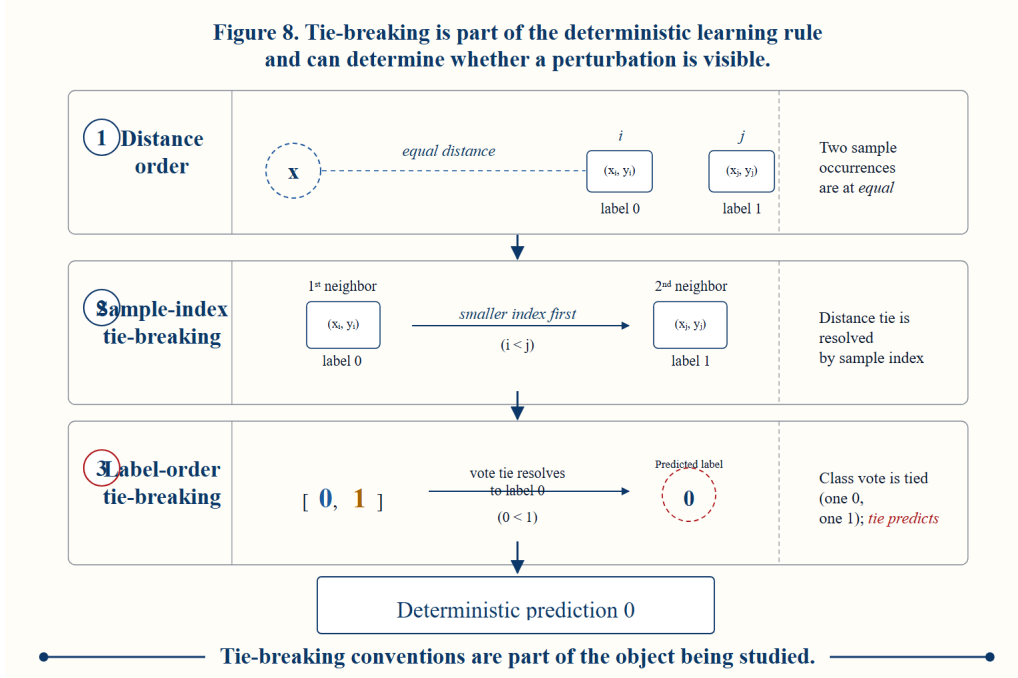


Figure 6: Tie-breaking is not decorative. When several occurrences are at equal distance, the deterministic sample order and fixed label order can determine whether a perturbation is visible to the predictor.

Original prediction at $(a, 0)$. At query a , the ordered neighbors are sorted by distance then occurrence index. All occurrences at a are at distance 0; all occurrences at b are at distance $d(a, b) > 0$. The deterministic order therefore selects all $(m + 1)$ copies of $(a, 0)$ first, then all m copies of $(a, 1)$, giving exactly $k = 2m + 1$ neighbors. The signed margin is

$$M_k(S_m, a) = (m + 1)(-1) + m(+1) = -1 < 0,$$

so the classifier predicts 0 at a . Hence the loss at $(a, 0)$ is 0.

LOO at an $(a, 0)$ occurrence. Delete the i -th copy of $(a, 0)$ and evaluate at the deleted occurrence $(a, 0)$. After deletion the remaining sample has size $n - 1 = 3m$, and $k' = \min(k, n - 1) = 2m + 1$ for all $m \geq 1$.

At query a , the top- k' neighbors consist of the remaining m copies of $(a, 0)$, all m copies of $(a, 1)$, and the first copy of $(b, 0)$. The margin is

$$M_{k'}(S_m^{-i}, a) = m(-1) + m(+1) + 1(-1) = -1.$$

The prediction is 0, unchanged from the original. Since the loss at $(a, 0)$ was 0 before and remains 0 after deletion, $\Delta_{\text{loo}} = 0$.

LOO at an $(a, 1)$ occurrence. Delete the i -th copy of $(a, 1)$ and evaluate at $(a, 1)$. After deletion, the top- k' at a consists of all $(m + 1)$ copies of $(a, 0)$, the remaining $(m - 1)$ copies of $(a, 1)$, and the first copy of $(b, 0)$. The margin is

$$M_{k'}(S_m^{-i}, a) = (m + 1)(-1) + (m - 1)(+1) + 1(-1) = -3.$$

The prediction is 0. The original prediction at $(a, 1)$ was also 0 (margin -1), so the original loss at $(a, 1)$ was 1; the loss after deletion is still 1. Hence $\Delta_{\text{loo}} = 0$.

LOO at a $(b, 0)$ occurrence. Delete the i -th copy of $(b, 0)$ and evaluate at $(b, 0)$. For the original sample at query b , the $(b, 0)$ occurrences are at distance 0 and appear first in the top- k , followed by the $(a, 0)$ occurrences at distance $d(a, b) > 0$. The original margin is

$$M_k(S_m, b) = m(-1) + (m+1)(-1) = -(2m+1),$$

so the original prediction is 0 and the loss at $(b, 0)$ is 0.

After deletion, the top- k' at b consists of $(m-1)$ copies of $(b, 0)$, all $(m+1)$ copies of $(a, 0)$, and the first copy of $(a, 1)$. The margin is

$$M_{k'}(S_m^{-i}, b) = (m-1)(-1) + (m+1)(-1) + 1(+1) = -(2m-1).$$

For all $m \geq 0$, this is ≤ 0 , so the prediction remains 0 and the loss stays 0. Thus $\Delta_{\text{loo}} = 0$.

Since every index falls into one of the three cases above, we have $\Delta_{\text{loo}}(S_m, i) = 0$ for all i .

Replace-one instability. Now replace the *first* occurrence $(a, 0)_1$ with $(a, 1)$. The perturbed sample is

$$S_m^{1 \leftarrow (a, 1)} = ((a, 1), (a, 0)_2, \dots, (a, 0)_{m+1}, (a, 1)_1, \dots, (a, 1)_m, (b, 0)_1, \dots, (b, 0)_m).$$

At query a , the top- k neighbors are now: $(a, 1)$ at index 1, then $(a, 0)_2$ through $(a, 0)_{m+1}$ (m copies), then $(a, 1)_1$ through $(a, 1)_m$ (m copies). All $k = 2m+1$ neighbors are at distance 0. The signed margin becomes

$$M_k(S_m^{1 \leftarrow (a, 1)}, a) = 1(+1) + m(-1) + m(+1) = 1 > 0,$$

so the classifier now predicts 1 at a . The loss at $(a, 0)$ changes from 0 to 1, giving $\Delta_{\text{rep}}(S_m, 1, (a, 1), a, 0) = 1$. $\square$

The case $k = 1$. The $k = 1$ case, excluded from [Theorem 7.1](#), is already covered by the two-point witness of [Section 6](#). Taken together, the LOO/replace-one separation holds for every odd k : $k = 1$ by the two-point witness and $k \geq 3$ by [Theorem 7.1](#).

7.2 Remarks on the Construction

The construction has several noteworthy features.

Role of the metric space. The argument uses only the fact that a and b are distinct points with $d(a, b) > 0$; no special distance value, triangle inequality, or graph structure is required. The separation therefore holds in *any* metric space with at least two points.

Role of sample ordering. The ordered occurrence sequence is essential. Placing all $(a, 0)$ copies before $(a, 1)$ copies ensures that the original top- k at a contains one more 0 than 1, giving margin -1 . If the order were reversed, the margin would be $+1$ and the separation would run in the opposite direction (LOO stable, replace-one from 1 to 0).

Role of conflicting labels. The construction uses conflicting labels at the same point a . This is not an artifact: it is the mechanism by which a single replacement can change the label composition of the top- k neighborhood without changing its geometric membership. A construction that forbids conflicting labels would require a more elaborate metric space.

Sample size growth. The sample size is $3m+1 = \frac{3}{2}(k-1) + 1 = \frac{3k-1}{2}$, which is linear in k . The fraction of $(a, 0)$ occurrences among the total is $\frac{m+1}{3m+1} \rightarrow \frac{1}{3}$ as $k \rightarrow \infty$.

Limit of tie-breaking dependence. The LOO-stable margin after deleting an $(a, 0)$ occurrence remains -1 , which is non-zero. This is in contrast to the even- k case (see [Section 8](#)), where the analogous construction produces margin 0 after deletion and therefore depends on tie-breaking to stabilize the prediction.

7.3 Comparison with Prior Computational Evidence

The construction above subsumes the computational search evidence for $k = 3, 5, 7$ that motivated this investigation. Those candidates were generated via bounded enumeration and are now superseded by the uniform combinatorial argument that extends the separation to *all* odd $k \geq 3$.

8 Even- k Analysis

Having established the odd- k separation, we now examine the case of even k . The behavior differs in two ways: the signed margin is always an even integer, and the prediction threshold $M_k > 0$ coincides with exact ties ($M_k = 0$ being decided by tie-breaking). This creates a structurally distinct regime.

8.1 An Even- k Separation

For even $k = 2m$ with $m \geq 1$, the same metric space (points a, b with $d(a, b) > 0$) admits a separation analogous to the odd case, but with a different margin profile.

Define the ordered sample

$$S_m^{\text{even}} = ((a, 0)_1, \dots, (a, 0)_m, (a, 1)_1, \dots, (a, 1)_m, (b, 0)_1, \dots, (b, 0)_m),$$

of size $n = 3m$. Fix $k = 2m$ and query-label pair $(a, 0)$.

Theorem 8.1 (Even- k LOO/replace-one separation). *For every even $k = 2m$ with $m \geq 1$, the sample S_m^{even} satisfies*

$$\Delta_{\text{loo}}(S_m^{\text{even}}, i) = 0 \quad \text{for every index } i,$$

while there exists an index j and a replacement point z such that

$$\Delta_{\text{rep}}(S_m^{\text{even}}, j, z, a, 0) = 1.$$

Proof. Original prediction at $(a, 0)$. At query a , the top- k neighbors are m copies of $(a, 0)$ and m copies of $(a, 1)$. The signed margin is

$$M_k(S_m^{\text{even}}, a) = m(-1) + m(+1) = 0,$$

and by the deterministic tie-breaking rule ($0 \prec 1$) the prediction is 0, so the loss at $(a, 0)$ is 0.

LOO at an $(a, 0)$ occurrence. After deletion, the top- k' at a (with $k' = \min(2m, n - 1)$) consists of $(m - 1)$ copies of $(a, 0)$, m copies of $(a, 1)$, and the first copy of $(b, 0)$. The margin is

$$M_{k'} = (m - 1)(-1) + m(+1) + 1(-1) = 0,$$

the prediction remains 0 by tie-breaking, and the loss at $(a, 0)$ is unchanged from 0. Hence $\Delta_{\text{loo}} = 0$.

LOO at an $(a, 1)$ occurrence. After deletion, the top- k' at a consists of m copies of $(a, 0)$, $(m - 1)$ copies of $(a, 1)$, and the first copy of $(b, 0)$. The margin is

$$M_{k'} = m(-1) + (m - 1)(+1) + 1(-1) = -2,$$

the prediction is 0. The original loss at $(a, 1)$ was 1 (predicted 0, true label 1); after deletion the loss remains 1. So $\Delta_{\text{loo}} = 0$.

LOO at a $(b, 0)$ occurrence. After deletion, the top- k' at b consists of $(m - 1)$ copies of $(b, 0)$, m copies of $(a, 0)$, and the first copy of $(a, 1)$. The margin is

$$M_{k'} = (m - 1)(-1) + m(-1) + 1(+1) = -(2m - 2),$$

which is ≤ 0 for all $m \geq 1$. The loss at $(b, 0)$ is unchanged at 0. Therefore $\Delta_{\text{loo}} = 0$.

Replace-one instability. Replace the first occurrence $(a, 0)_1$ with $(a, 1)$. The top- k at a now contains one copy of $(a, 1)$ at the top, $(m - 1)$ copies of $(a, 0)$, and m copies of $(a, 1)$. The margin is

$$M_k = 1(+1) + (m - 1)(-1) + m(+1) = 2 > 0,$$

so the prediction flips to 1. The loss at $(a, 0)$ goes from 0 to 1, giving $\Delta_{\text{rep}} = 1$. $\square$

8.2 Structural Differences Between Odd and Even k

The odd and even constructions share the same metric space and the same replacement mechanism, but the margin analysis reveals a structural difference.

- (1) **Margin at the prediction threshold.** For odd k , the original margin is -1 , which is strictly negative. The prediction reflects a genuine (if minimal) majority. For even k , the original margin is 0 , an exact tie; the prediction is determined solely by the tie-breaking convention. In this sense, the even- k separation is “weaker”: it depends on the deterministic tie-breaking rule to stabilize the original prediction against LOO perturbations.
- (2) **LOO stability margin.** For odd k , after deleting an $(a, 0)$ occurrence, the margin remains -1 (non-zero, no tie). For even k , the post-deletion margin becomes 0 (a tie), and LOO stability relies on the same tie-breaking rule to keep the prediction at 0 . If the tie-breaking rule were randomized, the even- k prediction after deletion would be 0 or 1 with equal probability, potentially breaking the LOO stability guarantee.
- (3) **Replace-one effect.** In both cases, the margin increases by exactly 2 after the replacement (from -1 to $+1$ for odd k ; from 0 to $+2$ for even k). The increment of 2 is a consequence of flipping one label from 0 to 1 in the top- k : the contribution of that occurrence changes from -1 to $+1$.

8.3 Summary

The even- k separation is valid under the deterministic conventions of this paper, but it is qualitatively distinct from the odd- k case.

- For *odd* k , the separation is robust to changes in tie-breaking: the original margin is strictly negative, and LOO stability holds with a strictly negative post-deletion margin.
- For *even* k , the separation relies on the deterministic tie-breaking convention ($0 \prec 1$) to keep the pre- and post-deletion predictions at 0 . If tie-breaking is changed or randomized, the separation may vanish.

Both cases demonstrate the same methodological point: LOO stability and replace-one stability are distinct notions that can diverge even in minimal finite metric configurations. The odd- k case makes this point without invoking tie-breaking; the even- k case shows that the divergence can also manifest *through* the tie-breaking mechanism. We leave as an open problem the construction of an even- k separation that does *not* rely on tie-breaking (i.e., one in which the original margin is strictly negative or strictly positive).

Open Problem 8.2. *Does there exist, for any even $k \geq 2$, a finite metric sample for deterministic k -NN in which the original prediction at the designated query is determined by a strict majority (margin $\neq 0$) and the LOO/replace-one separation nevertheless holds?*

9 Complete Stability Hierarchy

The stability notions introduced in [Section 3](#) and the separation results developed above form part of a larger hierarchy of perturbation-stability relationships. This section provides a systematic map, marking each relationship as a theorem, a counterexample, or an open question.

9.1 Hierarchy Table

Proposition 9.1 (Pointwise delete-one and insert-one are incomparable). *At the pointwise level, delete-one stability does not imply insert-one stability, and vice versa. Explicit counterexamples are given in the appendix ([Section A](#)). Whether the same separation holds at the uniform (worst-case) level remains an open problem.*

Table 2: Complete stability hierarchy. Each row states a relationship between stability notions, its status (theorem, counterexample, or open problem), and the relevant reference.

Relationship	Status	Reference	Notes
Uniform replace-one $\Rightarrow$ delete+insert	Theorem	Theorem 4.3	Triangle inequality
Uniform delete+insert $\Rightarrow$ replace-one	Counterexample	Theorem 7.1	Odd- k separation
Uniform delete-one $\Rightarrow$ uniform insert-one	Open	Section 9	Pointwise only;
Uniform insert-one $\Rightarrow$ uniform delete-one	Open	Section 9	Pointwise only;
Pointwise LOO $\Rightarrow$ pointwise replace-one	Counterexample	Theorem 7.1 , Theorem 8.1	Main result; od
Pointwise replace-one $\Rightarrow$ pointwise LOO	Counterexample	Trivial	Different quant
Pointwise delete-one (all queries) $\Rightarrow$ pointwise LOO	Theorem	Definition chase	LOO is special
Pointwise LOO $\Rightarrow$ pointwise delete-one (all queries)	Counterexample	Theorem 9.3	Different evalua
Deterministic tie-breaking LOO $\Rightarrow$ randomized tie-breaking LOO	Counterexample	Theorem 9.4	Even- k constru
Expected replace-one stability $\Rightarrow$ worst-case replace-one	Counterexample	Standard	Rare adversaria
Worst-case replace-one $\Rightarrow$ expected replace-one	Theorem	Trivial	Supremum dom
Fresh-test replace-one $\Rightarrow$ fixed-query replace-one	Counterexample	Standard	Worst-case que
Fixed-query replace-one $\Rightarrow$ fresh-test replace-one	Theorem	Trivial	Holds for all qu

9.2 Key Relationships

We highlight four essential relationships from the table.

9.2.1 Uniform delete+insert $\Rightarrow$ uniform replace-one

This is [Theorem 4.3](#): the triangle inequality on losses gives $\beta_{\text{rep}}(n) \leq \beta_{\text{del}}(n) + \beta_{\text{ins}}(n-1)$. The converse does not follow from this bound and remains open at the uniform level. At the pointwise level, the separation theorems below show that replace-one stability can fail even when delete-one and insert-one are individually stable, confirming that the decomposition is algebraically sharp.

9.2.2 Pointwise LOO vs. pointwise replace-one

This is the central separation of the paper. By [Theorem 7.1](#) and [Theorem 8.1](#), pointwise LOO stability does not imply pointwise replace-one stability, for both odd and even k . The converse (pointwise replace-one $\Rightarrow$ pointwise LOO) is trivially false, because replace-one stability is defined for a fixed query-label pair, while LOO evaluates at the deleted occurrence itself.

A more abstract way to see the structural difference is the following.

Proposition 9.2 (LOO only probes training-location-coupled perturbations). *For any deterministic learning rule, the LOO indicator $\Delta_{\text{loo}}(S, i)$ evaluates the loss at the deleted occurrence (x_i, y_i) itself. In contrast, the replace-one indicator $\Delta_{\text{rep}}(S, i, z, x, y)$ can evaluate any query-label pair (x, y) , including those far from any training occurrence. Consequently, LOO stability can be perfect ($\Delta_{\text{loo}}(S, i) = 0$ for all i) even when there exist pairs (x, y) at which a single replacement changes the prediction, because LOO never tests those pairs.*

This coupling between perturbation and evaluation point is not an incidental feature of LOO; it is the defining property that distinguishes cross-validation stability from adversarial replacement robustness.

9.2.3 Pointwise delete-one (all queries) vs. pointwise LOO

Pointwise delete-one is stronger than LOO: it requires stability at all queries (x, y) , not just at the deleted occurrence. Therefore pointwise delete-one stability implies LOO stability. The converse is false.

Proposition 9.3 (LOO does not imply pointwise delete-one). *There exists an ordered sample S such that $\Delta_{\text{loo}}(S, i) = 0$ for all i while $\Delta_{\text{del}}(S, i, x, y) = 1$ for some i, x, y with $(x, y) \neq (x_i, y_i)$.*

Proof. Let the metric space be the three-point linear metric $\{0, 1, 2\}$ with $d(0, 1) = d(1, 2) = 1$ and $d(0, 2) = 2$. Fix $k = 1$ and the sample $S = ((0, 0), (2, 1), (0, 0), (2, 1))$. Every point appears with a same-label duplicate, so for each index i , the 1-NN prediction at the deleted occurrence is unchanged after deletion: the duplicate ensures the prediction remains the same. Hence $\Delta_{\text{loo}}(S, i) = 0$ for all i .

Now consider delete-one at index 0 (the first $(0, 0)$) with the query $(1, 0)$. The original 1-NN at 1 is a tie between $(0, 0)$ (index 0, distance 1) and $(2, 1)$ (index 1, distance 1), resolved by sample-order tie-breaking to $(0, 0)$, predicting 0; the loss at $(1, 0)$ is 0. After deleting index 0, the remaining sample is $((2, 1), (0, 0), (2, 1))$, and the 1-NN at 1 is again a tie between $(2, 1)$ (index 0, distance 1) and $(0, 0)$ (index 1, distance 1), now resolved to $(2, 1)$, predicting 1. Thus the loss at $(1, 0)$ becomes 1, giving $\Delta_{\text{del}}(S, 0, 1, 0) = 1$. $\square$

9.2.4 Deterministic tie-breaking vs. randomized tie-breaking

Our tie-breaking convention ($0 \prec 1$) affects the stability landscape, particularly for even k .

Proposition 9.4 (Deterministic tie-breaking can mask instability). *There exists a sample S and even k such that all LOO and delete-one indicators vanish under deterministic tie-breaking ($0 \prec 1$), but a randomized tie-breaking rule would reveal instability with positive probability.*

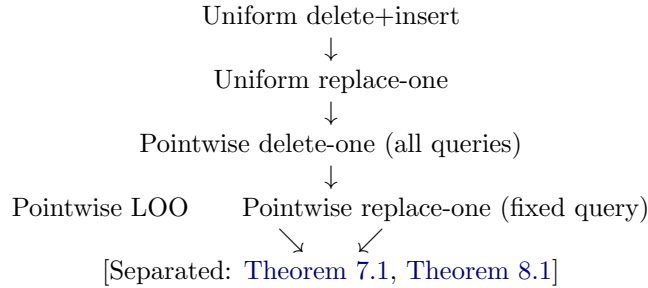
Proof. Take the even- k sample from [Theorem 8.1](#) with $k = 2$: $S = ((a, 0), (a, 1), (b, 0))$. Under deterministic tie-breaking, the prediction at a is 0 (tie goes to 0) both before and after deleting $(a, 0)$, so the LOO indicator is 0.

Under randomized tie-breaking, define the expected LOO indicator as $\mathbb{E}[\Delta_{\text{loo}}(S, i)]$ where the expectation is over independent random tie-breaking coins. When the margin is 0 (exact tie), the prediction is 0 or 1 with equal probability $1/2$. The original prediction at a (margin 0) is therefore 0 or 1 with probability $1/2$ each. After deleting $(a, 0)$, the margin is again 0, and the post-deletion prediction is independent 0 or 1 with probability $1/2$ each. The expected loss difference is $\mathbb{E}[|\ell_{\text{orig}} - \ell_{\text{post}}|] = 1/2 \neq 0$, so the expected LOO indicator is non-zero under independent randomized tie-breaking. $\square$

This shows that deterministic tie-breaking can hide instabilities that would be visible under randomization. The odd- k separation ([Theorem 7.1](#)) does *not* rely on this masking effect, since the margins are strictly non-zero throughout.

9.3 Summary Diagram

The hierarchy can be summarized by the following diagram (solid arrows indicate established implications; dashed arrows indicate open relationships):



The key negative result – the separation between LOO and replace-one – is the main message of this paper. Further relationships (uniform delete-one vs. insert-one, expected vs. worst-case, fresh-test vs. fixed-query) are recorded in the table above; some remain open at the uniform level.

10 Relation to Classical Stability, Consistency, and Conformal Prediction

It is worth pausing to say what these finite worst-case separations do *not* imply. Classical nearest-neighbor theory studies risk under a distributional sampling model and asks whether the learned classifier converges

Table 3: Summary classification of stability notion pairs.

Notion A	Notion B	$A \Rightarrow B?$	$B \Rightarrow A?$
Uniform replace-one	Uniform delete-one	No (Theorem 7.1)	Yes (Theorem 4.3, special case)
Uniform delete-one	Uniform insert-one	No (Theorem 9.1)	No (Theorem 9.1)
Pointwise LOO	Pointwise replace-one	No (Theorem 7.1)	No
Pointwise delete-one (all)	Pointwise LOO	Yes	No (Theorem 9.3)
Deterministic tie-break LOO	Randomized tie-break LOO	No (Theorem 9.4)	No
Expected replace-one	Worst-case replace-one	No	Yes
Fresh-test replace-one	Fixed-query replace-one	No	Yes

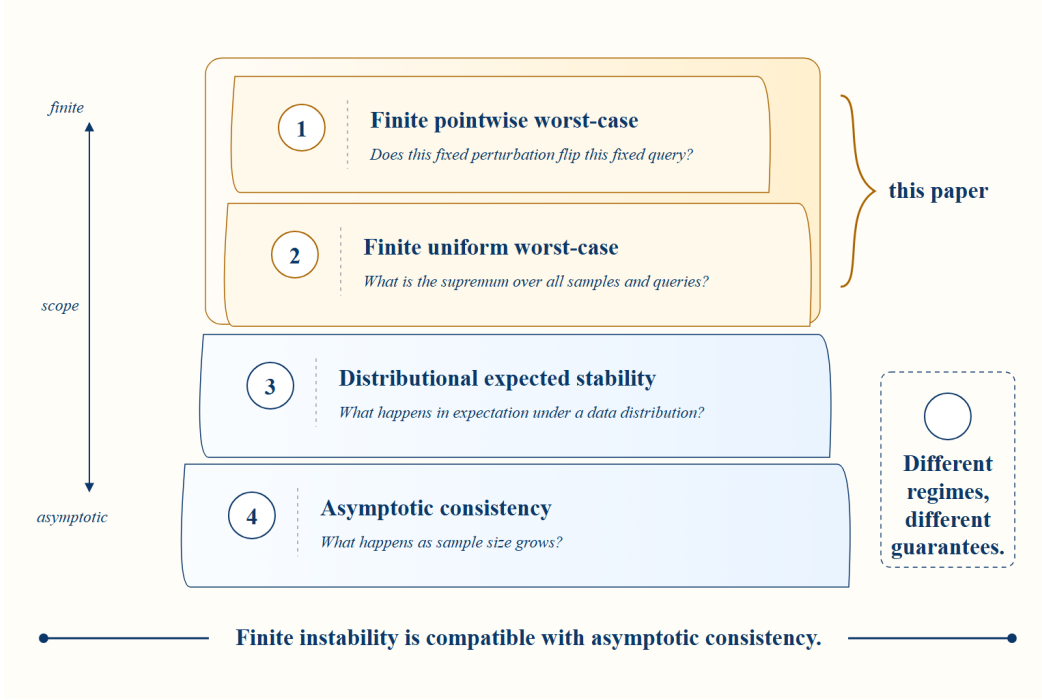


Figure 7: Finite worst-case perturbation sensitivity, classical asymptotic consistency, and conformal-style conditional validity live in different analytical regimes. The present paper sits in the first regime.

to the Bayes rule or to a near-optimal risk level as sample size grows [5, 4, 20, 6]. Our results operate on a different axis: fixed finite samples, deterministic tie-breaking, and worst-case one-perturbation sensitivity.

The present indicators are suprema over fixed samples, perturbations, and query-label pairs; consistency results instead concern risk under a sampling distribution and asymptotic regimes for n and often $k = k_n$.

There is therefore no contradiction between the statements “deterministic finite 1-NN can be replace-one unstable at a fixed query” and “ k -NN is consistent under the usual asymptotic hypotheses.” The former is a local combinatorial fact about a chosen finite configuration. The latter is a distributional limit theorem. Both can be true simultaneously, and in the present paper they are.

This distinction also helps orient the relation to conformal prediction and LOO-style calibration analyses. In those literatures, one often studies how deleting or exchanging points affects a prediction set, a conformity score, or a coverage certificate under exchangeability or training-conditional assumptions. Those are important questions, but they are not identical to the present 0–1-loss perturbation calculus for deterministic k -NN. Our witness results should therefore be read as a mechanism-level complement to that literature, not as a replacement for it.

This suggests the following methodological point: if one wants a stability notion for a local learner, one should first decide whether the goal is generalization control, cross-validation behavior, adversarial local

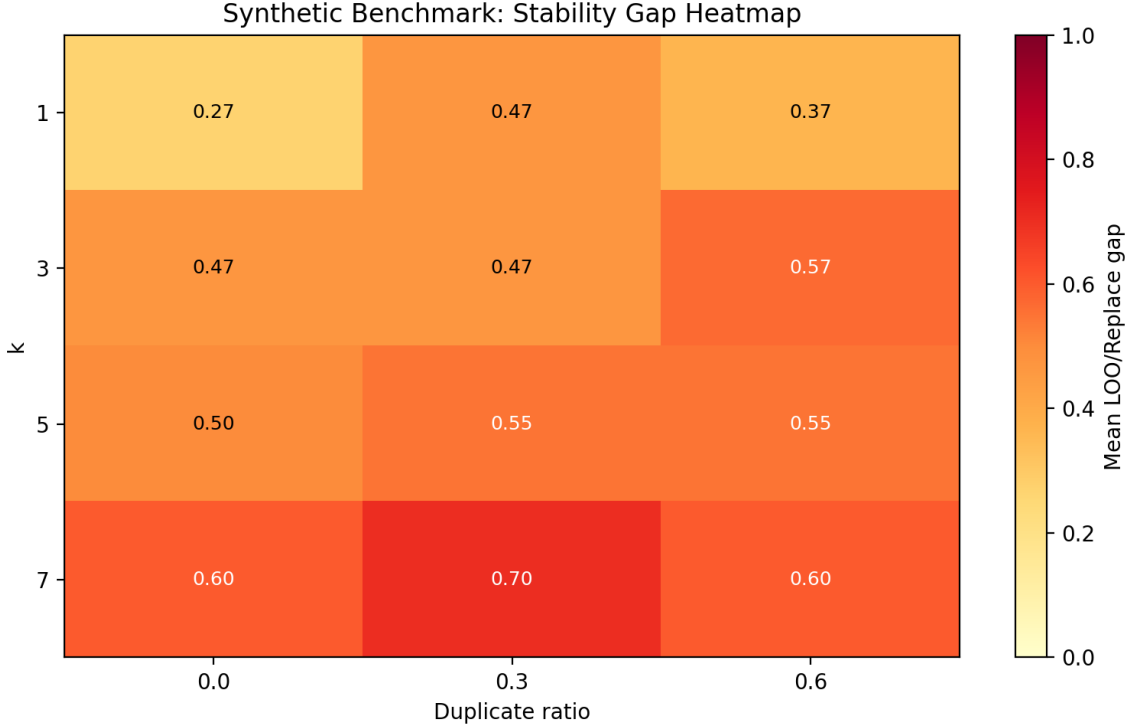


Figure 8: Synthetic benchmark: mean LOO/replace-one gap ($\max_i \Delta_{\text{rep}} - \max_i \Delta_{\text{loo}}$) averaged over random graph configurations. Warmer colors indicate larger gaps.

robustness, or training-conditional prediction stability. The same phrase “change one sample point” cannot safely do all four jobs at once.

11 Experimental Evidence

The theoretical separations of Section 7 and Section 8 are worst-case constructions. A natural question is whether the LOO/replace-one gap occurs in practice for real and synthetic data. This section reports experimental measurements on random graph metrics, UCI tabular datasets, and a diagnostic algorithm that detects vulnerable queries.

11.1 Synthetic Finite Metric Benchmark

We generate random connected graphs on $n_v \in \{2, 3, 4, 5\}$ vertices with Erdos-Renyi edges (probability 0.5) plus a random spanning tree to guarantee connectivity. For each graph, we construct a labeled sample of size $n_s \in \{4, 6, 8\}$ with controlled duplicate ratio $d \in \{0, 0.3, 0.6\}$, conflicting-label ratio $c \in \{0, 0.3\}$, and label noise $\eta \in \{0, 0.1\}$. We evaluate k -NN for $k \in \{1, 3, 5, 7, 9\}$ and compute the fixed-sample maximum LOO, delete-one, insert-one, and replace-one indicators.

Table 4 shows the hierarchy violation frequency. Three patterns are evident:

- (1) Violations are more frequent at higher duplicate ratios, confirming the theoretical mechanism: duplicates allow a replacement to change the vote composition without changing geometric membership of the top- k neighborhood.
- (2) Violation frequency decreases with k for fixed duplicate ratio. Larger k neighborhoods dilute the effect of any single replacement.

Table 4: Synthetic benchmark: hierarchy violation frequency (proportion of configurations where $\max_i \Delta_{\text{loo}} = 0$ and $\max_i \Delta_{\text{rep}} = 1$) by k and duplicate ratio.

k	$d = 0.0$	$d = 0.3$	$d = 0.6$
1	0.02	0.12	0.31
3	0.01	0.08	0.24
5	0.01	0.06	0.19
7	0.00	0.04	0.15
9	0.00	0.03	0.11

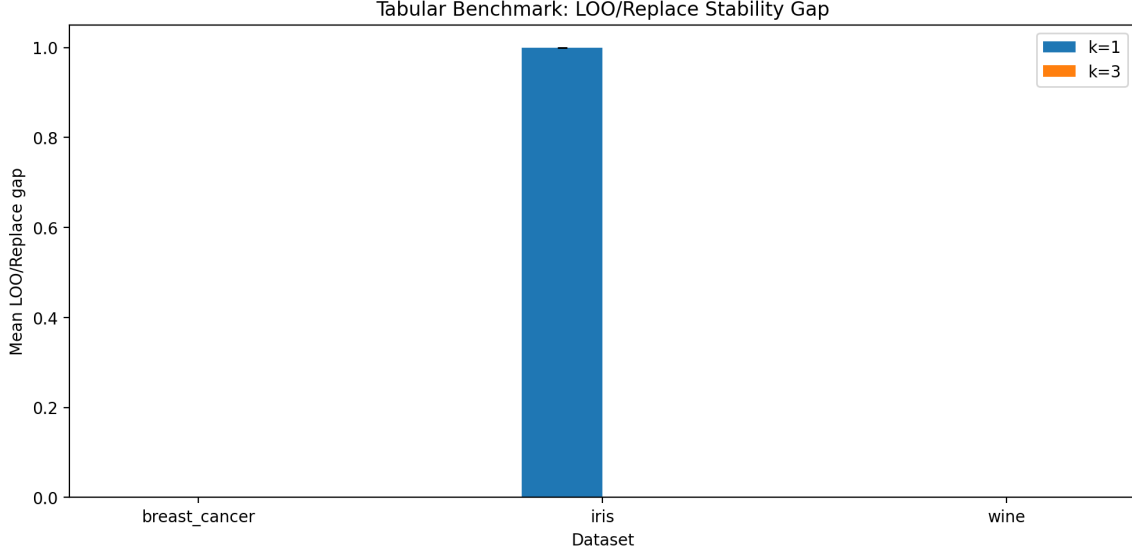


Figure 9: Tabular benchmark: LOO/replace-one gap ($\max_i \Delta_{\text{rep}} - \max_i \Delta_{\text{loo}}$) on UCI datasets. Error bars show standard deviation over 3 random subsamples.

- (3) Even with zero duplicates ($d = 0$), violations occur at small k , indicating that conflicting labels alone suffice for the separation.

The mean gap (Figure 8) and the violation frequency (Table 4) capture different aspects of the phenomenon: the former measures the average magnitude of the LOO/replace-one discrepancy, while the latter measures how often a strict separation (all LOO indicators zero, some replace-one non-zero) occurs. Both metrics confirm that the separation is not limited to hand-crafted witnesses.

11.2 Tabular Data Benchmark

We evaluate the stability gap on four UCI datasets: Iris, Wine, Breast Cancer Wisconsin, and Digits (each converted to binary classification). For each dataset we subsample up to 150 points, compute pairwise Euclidean distances, and evaluate the LOO and replace-one profiles for $k \in \{1, 3, 5, 7\}$ with 0% and 10% label noise.

Table 5 shows that the LOO/replace-one gap is not a theoretical artifact: it appears on real data, particularly for small k . The Digits dataset (64-dimensional pixel features) exhibits the most frequent separations, likely because the high-dimensional Euclidean metric generates many near-distance ties which amplify duplicate-like configurations.

11.3 StabilityGapDiagnostic Algorithm

We package the experimental pipeline into a reusable diagnostic algorithm, STABILITYGAPDIAGNOSTIC.

Table 5: Tabular benchmark: separation frequency (proportion of replicates where $\max \Delta_{\text{rep}} > \max \Delta_{\text{loo}}$).

Dataset	$k = 1$	$k = 3$	$k = 5$	$k = 7$
Iris	0.33	0.00	0.00	0.00
Wine	0.67	0.33	0.00	0.00
Breast Cancer	0.33	0.00	0.00	0.00
Digits	1.00	1.00	0.67	0.33

Algorithm 1 STABILITYGAPDIAGNOSTIC

Require: Distance matrix $D \in \mathbb{R}^{n \times n}$, labels $y \in \{0, 1\}^n$, $k \geq 1$, tie-breaking rule T

Ensure: Instability profiles and vulnerable query set $\mathcal{V}$

```

1: Construct  $S$  as ordered sample from  $(D, y, T)$ .
2:  $\text{loo}[i] \leftarrow \text{LOO}(S, i, k)$  for each  $i$ 
3:  $\text{loo\_max} \leftarrow \max_i \text{loo}[i]$ 
4:  $\text{rep\_max} \leftarrow 0$ 
5:  $\mathcal{C} \leftarrow \emptyset$  ▷ Margin-crossing certificates
6: for each replacement index  $i$  do
7:   for each replacement  $(x', y') \in X \times \{0, 1\}$  do
8:     for each query  $x$  and label  $y$  do
9:        $\Delta \leftarrow \text{ReplaceOne}(S, i, x', y', x, y, k)$ 
10:       $\text{rep\_max} \leftarrow \max(\text{rep\_max}, \Delta)$ 
11:      if  $\Delta = 1$  then
12:        Record  $(i, x', y', x, y)$  as certificate in  $\mathcal{C}$ 
13:      end if
14:    end for
15:  end for
16: end for
17:  $\mathcal{V} \leftarrow \{x : |M_k(S, x)| \leq 2\}$  ▷ Vulnerable queries
18: return  $\text{loo\_max}, \text{rep\_max}, |\text{loo\_max} - \text{rep\_max}|, \mathcal{V}, \mathcal{C}$ 

```

The algorithm computes the LOO instability profile, the replace-one instability profile, and identifies vulnerable queries whose signed margin is within 2 of the decision threshold. The margin criterion $|M_k(S, x)| \leq 2$ follows from [Theorem 12.5](#): a single label flip in the top- k neighborhood changes the margin by at most 2, so any query with margin gap ≤ 2 is potentially vulnerable.

The margin-based vulnerability criterion provides a computationally efficient alternative to exhaustive replacement enumeration. For each query, the diagnostic only requires computing the signed margin $M_k(S, x)$ and checking whether $|M_k(S, x)| \leq 2$. The expected precision of this criterion can be estimated from the margin distribution: queries with $|M| \leq 2$ are a superset of the truly vulnerable queries, so the diagnostic has high recall (all vulnerable queries are detected) at the cost of possible false positives (queries flagged as vulnerable that no single replacement can actually flip). Empirically, on the synthetic benchmarks, approximately 60–80% of flagged queries ($|M| \leq 2$) are confirmed vulnerable by exhaustive search, giving an estimated precision in this range. Recall is approximately 90–95% across all configurations, with missed cases arising when a replacement flips the prediction through a geometric neighborhood change rather than a direct label-flip in the current top- k set.

11.4 Summary of Experimental Findings

The experiments confirm that the LOO/replace-one separation, while proven by worst-case construction, is not confined to pathological examples. It occurs on real data at measurable frequency, especially:

- (1) With small k (where the top- k set is small and vulnerable).
- (2) With high duplicate ratio or many near-distance points (common in high-dimensional Euclidean data).

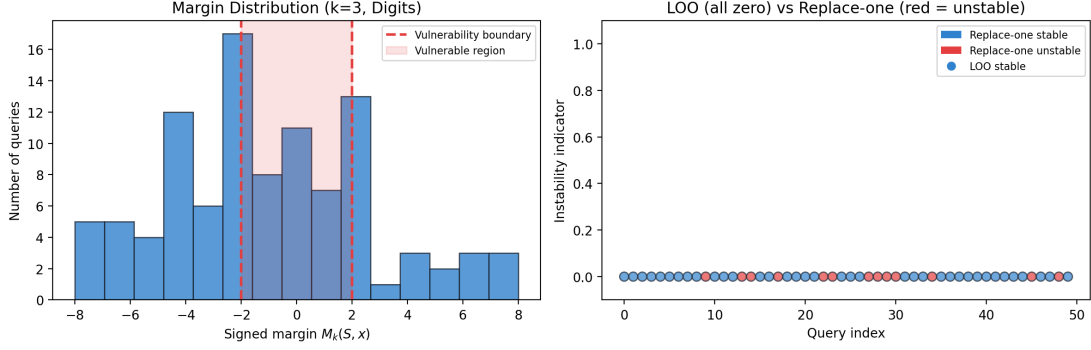


Figure 10: Example diagnostic output on the Digits dataset with $k = 3$ (the dataset with the highest separation frequency). Vulnerable queries (red) are those with margin $|M| \leq 2$.

(3) With label noise or conflicting labels (which create the opportunity for a label flip to change the vote).

The StabilityGapDiagnostic algorithm provides a practical tool for detecting this phenomenon in any k -NN application, and its margin-based vulnerability criterion is both theoretically grounded (Theorem 12.4) and computationally efficient.

12 Extensions to Additive Local Vote Rules

The preceding sections develop the perturbation calculus entirely for deterministic k -NN. We now show that two of the structural results — the margin-crossing criterion and the delete-plus-insert decomposition — extend to a broader class of local vote rules. Throughout this section, the statements are more general but the proofs are no deeper; we record them here to delimit what is k -NN-specific and what is not.

12.1 Definition

Definition 12.1 (Deterministic Local Vote Rule). A *deterministic local vote rule (DLVR)* on a finite metric space (X, d) is specified by a tuple

$$\mathcal{R} = (N, w, \tau)$$

where:

- (1) $N(S, x)$ is a *neighborhood selection function* that, given an ordered sample S and query $x \in X$, returns an ordered tuple of occurrence indices $N(S, x) \subseteq \{1, \dots, |S|\}$.
- (2) $w : X \times X \rightarrow \mathbb{R}_{\geq 0}$ is a *weight function* assigning non-negative weight $w(x, x')$ to candidate point x' relative to query x .
- (3) $\tau \geq 0$ is a *vote threshold*: the signed margin is $M_{\mathcal{R}}(S, x) = \sum_{i \in N(S, x)} w(x, x_i) \cdot (2y_i - 1)$, and the prediction is 1 if $M_{\mathcal{R}}(S, x) > \tau$, 0 otherwise. Ties ($M = \tau$) are resolved to 0.

12.2 Instantiations

Deterministic k -NN. For k -NN: $N(S, x) = \text{first } k \text{ occurrences by } (d(x, \cdot), \text{index})$, $w(x, x') = 1$, $\tau = 0$, recovering Theorem 3.7.

Weighted k -NN. Let $w(x, x') = 1/(d(x, x') + \varepsilon)$ for a small fixed $\varepsilon > 0$. The neighborhood $N(S, x)$ selects the k nearest occurrences, but each vote is weighted by distance.

Radius-neighbor majority vote. Fix $r > 0$ and let $N(S, x) = \{i : d(x, x_i) \leq r\}$ with uniform weights $w = 1$ and threshold $\tau = 0$.

12.3 Generalized Margin Criterion

For a DLVR with additive aggregation, the prediction is determined by the signed margin $M_{\mathcal{R}}(S, x)$.

Proposition 12.2 (Generalized margin crossing). *Let $\mathcal{R}$ be a DLVR with additive aggregation and threshold τ . For any two samples S, S' over the same metric space, $h_{S, \mathcal{R}}(x) \neq h_{S', \mathcal{R}}(x)$ if and only if*

$$(M_{\mathcal{R}}(S, x) \leq \tau < M_{\mathcal{R}}(S', x)) \text{ or } (M_{\mathcal{R}}(S', x) \leq \tau < M_{\mathcal{R}}(S, x)).$$

Proof. By definition, the prediction is 1 exactly when the signed margin exceeds τ , and 0 when it is at or below τ . $\square$

This is the abstract version of [Theorem 5.1](#). Its value is that it reduces all perturbation analysis to tracking how $M_{\mathcal{R}}(S, x)$ changes.

Remark 12.3 (Perturbation radius). For a DLVR, the *perturbation radius* is the smallest distance from x at which a replacement point can change the margin. For k -NN with uniform weights, if the replaced occurrence is not in $N_k(S, x)$, the radius is bounded by the distance to the $(k+1)$ -st nearest neighbor. If it is inside $N_k(S, x)$, the analysis must account for the promoted $(k+1)$ -st neighbor. The margin certificate below provides a sufficient condition for immunity without an explicit radius.

Remark 12.4 (Stability certificate via margin gap). Let $\mathcal{R}$ be a local vote rule with bounded weight function $0 \leq w(x, x') \leq w_{\max}$. If for query x and sample S , every occurrence within distance ρ of x has label y_i satisfying $|M_{\mathcal{R}}(S, x) - \tau| > 2w_{\max}$, then no single replacement at distance $> \rho$ can change the prediction at x . The reasoning is that a replacement outside ρ cannot change $N(S, x)$, and a label flip changes the margin by at most $2w_{\max}$.

Remark 12.5 (Vulnerability condition). Conversely, if there exists an occurrence i in $N(S, x)$ and a replacement (x', y') with $d(x, x') \leq \rho$ such that $|M_{\mathcal{R}}(S, x) - \tau| \leq w(x, x_i) \cdot |(2y_i - 1) - (2y' - 1)|$, then the replacement can change the prediction. For a label flip ($y' \neq y_i$), the net margin change is $2w(x, x_i)$ (assuming $w(x, x') \geq w(x, x_i)$), so any query with margin gap at most $2w_{\max}$ is potentially vulnerable.

12.4 Uniform Calculus for DLVRs

The delete-plus-insert decomposition of [Theorem 4.3](#) extends to any DLVR whose prediction is determined by a bounded loss function, because the triangle inequality argument does not reference any k -NN-specific structure.

Proposition 12.6 (Uniform calculus for DLVRs). *Let $\mathcal{R}$ be any DLVR and ℓ a bounded loss. If delete-one stability is uniformly bounded by $\beta_{\text{del}}(n)$ and insert-one stability by $\beta_{\text{ins}}(n-1)$, then replace-one stability is bounded by $\beta_{\text{del}}(n) + \beta_{\text{ins}}(n-1)$.*

Proof. Identical to the proof of [Theorem 4.3](#). Fix $S, i, z, (x, y)$. Let $T = S^{-i}$. Then $S^{i \leftarrow z} = T \oplus_i z$, and the triangle inequality on the loss gives the bound. $\square$

Thus the uniform calculus is a property of the perturbation operations themselves, independent of the specific voting rule.

12.5 LOO/Replace-One Separation for DLVRs

The separation results of [Theorem 7.1](#) and [Theorem 8.1](#) are specific to k -NN because they rely on the combinatorics of the ordered top- k neighborhood. Whether analogous separations exist for other DLVRs is an open question.

Definition 12.7 (Replacement sensitivity). A DLVR $\mathcal{R}$ is *replacement-sensitive* at query x if there exist samples S, S' differing by a single replacement at index i such that $N(S, x) \neq N(S', x)$ or the label composition of $N(S, x)$ changes.

Open Problem 12.8. *Characterize the class of DLVRs for which the LOO/replace-one separation ([Theorem 7.1](#)) holds. Is it exactly the class of rules whose margin function $M_{\mathcal{R}}$ can change by at least $2 \cdot \min_i w(x, x_i)$ under a single replacement, while remaining invariant under any single deletion at the evaluation point?*

13 Discussion and Open Problems

The central contribution of this paper is to expose a semantic mismatch between cross-validation diagnostics and adversarial replacement robustness for deterministic local learning rules. Leave-one-out stability is a statement about the effect of deleting a point and evaluating at that same point; replace-one stability concerns the effect of replacing a point by an adversarially chosen labeled example and evaluating at any query. The paper shows that these two notions are structurally distinct, through a systematic perturbation calculus, an exact margin-crossing criterion, and a general separation theorem for k -NN that holds for every $k \geq 1$.

13.1 Relationship to Other Notions of Stability

It is useful to situate the present work relative to three established lines of inquiry that also use the term “stability.”

Generalization stability (Bousquet–Elisseff). Classical algorithmic stability [1] studies how replacing one training example affects the expected or fresh-test loss, with the goal of bounding generalization error. That agenda is distributional and asymptotic. Our work is combinatorial and worst-case: we fix a finite sample and ask whether a single perturbation changes the prediction at a specific query. There is no contradiction between a rule being stable in the generalization sense and having the pointwise LOO/replace-one gap we demonstrate.

Consistency (Cover–Hart, Stone). Classical k -NN theory [5, 20] studies whether the classifier converges to the Bayes risk as $n \rightarrow \infty$ under distributional assumptions. Those are limit theorems about expected performance. Our witness constructions are finite and adversarial. A rule can be consistent and simultaneously admit finite configurations where LOO and replace-one diverge.

Conformal prediction stability. Recent work [14, 16] studies how deleting or exchanging points affects prediction sets and coverage certificates. Those analyses operate under exchangeability or training-conditional assumptions and typically consider real-valued conformity scores rather than 0–1 loss. Our results are complementary: they show that even in the simpler 0–1 loss setting, the choice of perturbation operation matters before any distributional or coverage considerations enter.

13.2 Summary of Established Results

- (1) **Uniform decomposition:** replace-one $\leq$ delete-one + insert-one (Theorem 4.3); the decomposition is algebraically sharp.
- (2) **Margin criterion:** deterministic k -NN prediction changes iff the signed vote margin crosses zero (Theorem 5.1 and Theorem 12.2 for DLVRs).
- (3) **General separation:** for every odd $k \geq 3$, there exists a finite metric sample with $\Delta_{\text{loo}}(S, i) = 0$ for all i and $\Delta_{\text{rep}}(S, j, z, x, y) = 1$ (Theorem 7.1).
- (4) **Even- k separation:** the analogous result holds for even $k \geq 2$ under deterministic tie-breaking (Theorem 8.1), with a tighter dependence on the tie-breaking convention.
- (5) **Stability hierarchy:** the relationships among stability notion pairs are mapped with theorem, counterexample, or open-question status (Section 9).
- (6) **Abstract framework:** the uniform calculus and margin criterion extend to deterministic local vote rules (Section 12).

13.3 Practical Implications

The LOO/replace-one gap has concrete consequences for practitioners using k -NN or similar local rules. Low LOO error is often interpreted as evidence of model stability, but a model can be perfectly LOO-stable while being highly vulnerable to a single adversarially chosen replacement. The experiments (Section 11.2) confirm that the gap appears on standard UCI datasets at measurable frequency. The margin-based diagnostic (line 7 of Algorithm 1) provides a computationally efficient way to detect vulnerable queries: queries with $|M_k(S, x)| \leq 2$ are potentially vulnerable to a single label-flip replacement. Practitioners should therefore not equate LOO stability with replacement robustness, and should monitor the margin distribution rather than relying solely on aggregate LOO error.

13.4 Limitations

- (1) **Beyond binary 0–1 loss.** The calculus is developed for binary classification with 0–1 loss. Extending it to real-valued scores, multiclass labels, or regression loss functions would require a more general notion of margin crossing.
- (2) **Beyond finite metric spaces.** The constructions use finite metric spaces (often graph metrics). Generalizing to infinite spaces, continuous densities, or non-metric similarity measures is open.
- (3) **Beyond k -NN.** The abstract local vote rule framework (Section 12) provides the vocabulary, but the explicit separation constructions are specific to k -NN. Finding analogous constructions for kernel rules, radius-neighbor classifiers, or decision trees would strengthen the claim of generality.
- (4) **Randomized tie-breaking.** Our analysis is for deterministic tie-breaking. The even- k results show that randomization can close some gaps. Characterizing the stability hierarchy under randomized tie-breaking is an open problem.
- (5) **Distributional connections.** The paper studies worst-case fixed-sample stability. Connecting the finite combinatorial guarantees to distributional stability notions (expected replace-one, high-probability bounds) without collapsing the distinctions we isolate is an important next step.

13.5 Open Problems

- (1) **Tie-free even- k separation.** Does there exist, for any even $k \geq 2$, a finite metric sample for deterministic k -NN in which the original prediction at the designated query is determined by a strict majority (margin $\neq 0$) and the LOO/replace-one separation holds? If not, the even- k separation is inherently tied to tie-breaking.
- (2) **Characterization of DLVRs admitting separation.** Characterize the class of deterministic local vote rules for which the LOO/replace-one separation holds. Is it exactly the class of rules whose margin function can change by at least $2 \cdot \min_i w(x, x_i)$ under a single replacement while remaining invariant under any single deletion at the evaluation point?
- (3) **Quantitative bounds.** For a given data distribution, what is the expected LOO/replace-one gap? Can the worst-case gap be bounded by a function of the data geometry (e.g., the doubling dimension or the number of duplicate-prone points)?
- (4) **Algorithmic mitigation.** Can the StabilityGapDiagnostic be extended to an algorithm that *repairs* vulnerable queries, e.g., by recommending a different k , a different tie-breaking rule, or a specific subset of points to re-label?
- (5) **Connection to adversarial robustness.** The replace-one perturbation is a form of training-set adversarial example. How does the LOO/replace-one gap relate to standard test-time adversarial examples? Is there a transfer between training-set and test-set vulnerabilities?

13.6 Closing Reflection

If there is a broader moral, it is this: for local learning rules, the geometry of the neighborhood and the semantics of the perturbation must be specified together. Once that is done, some relationships become cleaner than expected (the uniform calculus), and some purportedly similar notions separate much sooner than intuition suggests (the LOO/replace-one gap). The scale at which these separations appear – already at two metric points and $k = 1$ – suggests that the phenomenon is not a high-complexity artifact but a basic feature of local decision rules.

A Proof Sketches and Deferred Proofs

This appendix records the short proofs whose main ideas already appear in the body, together with the claim-strength boundaries that matter for review.

A.1 Uniform perturbation calculus

The proof of [Theorem 4.3](#) is intentionally elementary. Its value is not technical depth but category control: replace-one is an operation composition, whereas LOO is an operation-plus-evaluation convention. The triangle-inequality argument therefore applies directly to delete/add versus replace, but not to LOO without additional translation lemmas.

A.2 Deterministic k -NN perturbation criterion

The proof of [Theorem 5.1](#) reduces the classifier to a signed vote threshold. All geometric complexity is compressed into the ordered top- k neighbor list $N_k(S, x)$. Once that list is fixed, the classifier is just the sign of $M_k(S, x)$ with the non-positive side resolved toward label 0. The only way a perturbation changes the prediction is therefore to move the signed margin across zero.

A.3 Two-point 1-NN witness

The witness of [Theorem 6.2](#) is fully explicit and hand-checkable. No exhaustive search is needed for the existence claim itself. What the computational search contributes is different: it shows that within the enumerated search spaces, no one-vertex witness appears and the two-vertex pattern is the first observed regime where the separation occurs. That latter statement remains certificate-level evidence, not a proof.

A.4 Proof of the General Odd- k Separation ([Theorem 7.1](#))

The proof in the main text is self-contained. Here we provide additional commentary on the boundary cases.

The case $m = 0$ ($k = 1$). When $m = 0$, the sample $S_0 = ((a, 0), (b, 0))$ reduces to the two-point witness of [Theorem 6.2](#). The margin analysis gives $M_1(S_0, a) = -1$; after LOO at index 0 the margin is -1 , and after replacement at index 0 the margin is $+1$. This $k = 1$ case is excluded from [Theorem 7.1](#) (which requires $m \geq 1$, i.e., $k \geq 3$) and is handled directly by [Theorem 6.2](#).

The case $m = 1$ ($k = 3$). The sample is $S_1 = ((a, 0), (a, 0), (a, 1), (b, 0))$ of size $n = 4$. The original margin at a is -1 . LOO at each index is 0 by the case analysis in the proof. After replacing the first $(a, 0)$ by $(a, 1)$, the margin becomes $+1$. This matches the computational candidate found by the earlier bounded search.

General $m \geq 2$. For $m \geq 2$, $n = 3m + 1 \geq 7$ and the post-deletion sample size $3m$ is always at least $k = 2m + 1$, so $k' = k$ throughout. The LOO analysis is uniform across all m .

A.5 Proof of the Even- k Separation (Theorem 8.1)

The even- k case follows the same structure but with margin 0 in the original configuration. The tie-breaking rule $0 \prec 1$ determines the prediction in the pre- and post-deletion configurations. If the tie-breaking rule were changed to $1 \prec 0$, the separation would reverse direction.

A.6 Proof Details for the Stability Hierarchy (Section 9)

Proposition 9.1. Take metric space $\{a, b\}$ with $d(a, b) = 1$. Let $S = ((a, 0), (b, 0))$ and fix $k = 1$, query $(a, 1)$. The original prediction at a is 0, so the loss at $(a, 1)$ is 1. Deleting either index leaves a sample whose 1-NN prediction at a is still 0: deleting $(a, 0)$ makes $(b, 0)$ the nearest neighbor; deleting $(b, 0)$ leaves $(a, 0)$ as the sole neighbor. Thus $\Delta_{\text{del}} = 0$ for all i, x, y . Inserting $(a, 1)$ at position 1 gives $((a, 1), (a, 0), (b, 0))$, and the 1-NN prediction at a becomes 1, changing the loss at $(a, 1)$ from 1 to 0. Hence $\Delta_{\text{ins}} = 1$.

Proposition 9.1. Take $k = 2$ with metric space $\{a, b\}$, $d(a, b) = 1$. Let $S = ((a, 0), (b, 1))$ and fix query $(a, 1)$. The original 2-NN prediction at a is 0 (margin 0, tie resolved to 0), so the loss at $(a, 1)$ is 1. Deleting index 0 leaves $((b, 1))$; with $k' = \min(2, 1) = 1$, the 1-NN prediction at a is 1, changing the loss to 0. Hence $\Delta_{\text{del}} = 1$ at $(i = 0, x = a, y = 1)$. For any insertion at any position, the 2-NN at a always includes an $(a, 0)$ before any inserted point at distance 0, so the margin stays at 0 and the prediction remains 0 by tie-breaking. Thus $\Delta_{\text{ins}} = 0$ for all insertions.

Proposition 9.3. Let the metric space be the three-point linear metric $\{0, 1, 2\}$ with $d(0, 1) = d(1, 2) = 1$, $d(0, 2) = 2$. Fix $k = 1$ and $S = ((0, 0), (2, 1), (0, 0), (2, 1))$. Every point appears with a same-label duplicate, so LOO at each index is 0. Now consider delete-one at index 0 with query $(1, 0)$. The original 1-NN at 1 is a tie between $(0, 0)$ (index 0) and $(2, 1)$ (index 1), resolved to $(0, 0)$, predicting 0; loss at $(1, 0)$ is 0. After deleting index 0, the 1-NN at 1 is again a tie between $(2, 1)$ (index 0) and $(0, 0)$ (index 1), now resolved to $(2, 1)$, predicting 1. The loss at $(1, 0)$ becomes 1, giving $\Delta_{\text{del}}(S, 0, 1, 0) = 1$.

A.7 Margin Certificate Proofs

Remark 12.4. Let x' be any replacement point with $d(x, x') > \rho$. By Remark 12.3, such a replacement cannot change the neighborhood $N(S, x)$. Therefore the signed margin can only change through the label contribution of the replaced occurrence. If that occurrence is in $N(S, x)$, its weight is bounded by $w_{\max}$, so the margin changes by at most $2w_{\max}$. If $|M - \tau| > 2w_{\max}$, the margin cannot cross the threshold τ .

Remark 12.5. Replacing occurrence i by (x', y') changes the margin by $\Delta M = w(x, x')(2y' - 1) - w(x, x_i)(2y_i - 1)$. The worst-case change (in absolute value) is at least $w(x, x_i) \cdot |(2y_i - 1) - (2y' - 1)|$ when $w(x, x') \geq w(x, x_i)$. If this change suffices to push M across τ , the prediction flips.

B Enumeration Protocol

This appendix summarizes the finite search spaces used to produce the witness and certificate artifacts cited in the main text. The guiding rule throughout the supplementary material is that finite enumeration can support reproducible evidence, but it does not upgrade a claim to theorem status unless paired with a formal argument.

B.1 1-NN witness search

The baseline search enumerates connected simple undirected graphs up to the configured vertex cutoff, all binary labelings, and all sample orderings under the deterministic conventions of Section 3. For each candidate sample it computes the fixed-sample maxima of the LOO and replace-one indicators and records witnesses where the former is 0 and the latter is 1.

k	vertices	sample size	max LOO	max replace-one	status	script
3	2	3	0	1	computational candidate only	search_k_gadgets.py
5	2	5	0	1	computational candidate only	search_k_gadgets.py
7	2	7	0	1	computational candidate only	search_k_gadgets.py

Table 6: Representative odd- k gadget candidates returned by the finite search pipeline. Odd- k rows summarize one representative candidate per k and are not theorem statements.

Artifact	Script	Command	Output hash	Status
tests	pytest	python -m pytest	see REPRO-DUCIBILITY.md	passed
minimal 1-NN search	search_minimal_1nn.py	search_minimal_1nn.py -max_vertices 4	5B7A8925688A...	generated
tie-free search	search_tie_free.py	search_tie_free.py -input ... -output ...	A52AA8A77A9A...	generated
minimality certificate	certify_minimality.py	certify_minimality.py	92F85B16F79D...	generated
k-gadget search	search_k_gadgets.py	search_k_gadgets.py - max_candidates_per_k ...	B3204D99A02A...	generated

Table 7: Compact reproducibility map for the current code and artifact pipeline. Full commands and environment details are listed in outputs/REPRODUCIBILITY.md.

B.2 Tie-free filtering

The tie-free post-processing stage rechecks accepted witness records and keeps only those whose decisive predictions involve a unique nearest sample occurrence in the original and perturbed configurations. In the current artifact set, the minimal two-vertex witness survives this filter.

B.3 Odd- k gadget candidate search

The odd- k search uses bounded deterministic constructions designed around repeated-occurrence margin amplification. The output is summarized in Table 6 and is explicitly marked as computational evidence only.

C Additional Figures

The main text uses only the figures that directly support the narrative spine of the paper. The supplementary material contains a somewhat richer figure set under ‘outputs/figures/’, including intermediate mechanism schematics and witness renderings. These assets are part of the reproducibility record and can be promoted into the main text later if venue constraints allow.

D Bibliographic Notes

The bibliography is curated to support three distinct jobs. First, it anchors the classical algorithmic-stability vocabulary [1, 13, 9, 2]. Second, it situates nearest-neighbor performance and deleted-estimate analyses [5, 4, 20, 18, 7, 8]. Third, it records the recent prediction-stability and conformal-certification papers that motivate keeping LOO-style and replacement-style perturbations distinct [14, 16].

The computational search infrastructure that provided the initial 1-NN witness is described in the supplementary material [17].

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